

Limits and Continuity

Faruk Uygul

University of Alberta

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Limits

We now study the continuity of functions with two variables. Consider a function $f(x, y)$ and $(a, b) \in \mathbb{R}^2$. We say that the limit of f at (a, b) is some number L if the values of $f(x, y)$ gets arbitrarily close to L as (x, y) approaches (a, b) .

In mathematical terms, we can define the limit as follows:

Definition We say that the limit of $f(x, y)$ as (x, y) approaches (a, b) is L if for every number $\epsilon > 0$ there is a corresponding number $\delta > 0$ such that

$$0 < \sqrt{(x - a)^2 + (y - b)^2} < \delta \implies |f(x, y) - L| < \epsilon$$

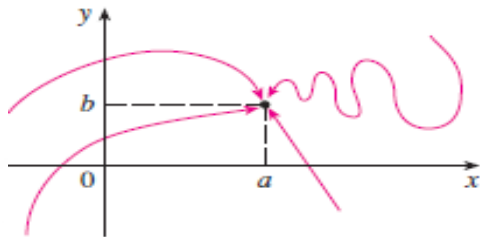
Notation

$$\lim_{(x, y) \rightarrow (a, b)} f(x, y) = L.$$

Recall that for $y = f(x)$, $\lim_{x \rightarrow a} f(x)$ exists if and only if $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$ exist and $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x)$.

This was because, in $\mathbb{R}$ one can approach a point either from right or left.

However, things are a bit more complicated in $\mathbb{R}^2$.



Suppose that $f(x, y) \rightarrow L_1$ as $(x, y) \rightarrow (a, b)$ along a path C_1 and $f(x, y) \rightarrow L_2$ as $(x, y) \rightarrow (a, b)$ along a path C_2 . If $L_1 \neq L_2$, then $\lim_{(x,y) \rightarrow (a,b)} f(x, y)$ does not exist.

This phenomena is sometimes called as **path independence of the limit**.

Definition We say that a function $f(x, y)$ is continuous at (a, b) if

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b).$$

Every polynomial, exponential function is continuous. Trigonometric functions, and logarithmic functions are continuous in their domains. Rational functions are continuous when their denominators are nonzero. Furthermore, a sum, difference, product or ratio* of continuous functions is continuous.

A polynomial in two variables is a linear combination of the terms of the form $cx^m y^n$ where $c \in \mathbb{R}$ and $n, m \in \mathbb{N}$.

Example 1. Evaluate the following limits, if they exist.

- $\lim_{(x,y) \rightarrow (3,-2)} (2x^2y - 5xy^3 + 41)$
- $\lim_{(x,y) \rightarrow (1,-1)} (e^{x-y} - 2 \cos(3x + y) + \sin(2y))$

Example 2. For the following limits approach $(0,0)$ along the x -axis, and along the y -axis, and along a line passing through the origin. Does the limit exist?

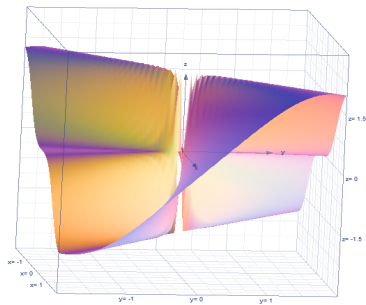
►
$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2}{x^2 + y^2}$$

► $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + \sin^2 y}{2x^2 + y^2}$

► $\lim_{(x,y) \rightarrow (0,0)} \frac{xy \cos y}{3x^2 + y^2}$

► $\lim_{(x,y) \rightarrow (0,0)} \frac{6x^3y}{2x^4 + y^4}$

Note that the graph of $z = \frac{6x^3y}{2x^4+y^4}$ looks like:



Example 3. Find

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + y^4}.$$

Example 4. Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2 + y^2}}$

Example 4. cont'd Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 \sin^2 y}{x^2 + 2y^2}$

Example 4. cont'd Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{4x^2y}{x^2 + y^2}$

Example 4. cont'd Evaluate

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y \cos\left(\frac{2}{x^3}\right)}{x^2 + 5y^2}$$

Example 5. Evaluate $\lim_{(x,y,z) \rightarrow (0,0,0)} \frac{x^2 + 2y^2 + 3z^2}{x^2 + y^2 + z^2}.$

Example 6. Determine the value(s) of c so that the following function is continuous at $(0,0)$.

$$f(x,y) = \begin{cases} \frac{x^2y^3}{2x^2+y^2}, & \text{if } (x,y) \neq (0,0) \\ c-1, & \text{if } (x,y) = (0,0) \end{cases}$$

Example 7. Determine whether the given function is continuous at $(0,0)$ or not.

$$f(x, y) = \begin{cases} \frac{xy}{x^2+xy+y^2}, & \text{if } (x, y) \neq (0, 0) \\ 0, & \text{if } (x, y) = (0, 0) \end{cases}$$

Example 8. Determine the set of points at which the function is continuous.

$$f(x, y) = \frac{\cos(x - y)}{e^x - y^2}, \quad g(x, y) = \ln(2 + 3x^2 + 4y^6) + \sqrt{x + y^2}$$

Example 9. Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^5 + y^5}{x^4 + y^4} = 0$.

Hint. Write

$$\frac{x^5 + y^5}{x^4 + y^4} = \frac{x^4 x + y^4 y}{x^4 + y^4} = \frac{x^4 x}{x^4 + y^4} + \frac{y^4 y}{x^4 + y^4}$$

Then, split the given limit into two limits, and use the squeeze theorem for each one.

We may want to try **polar coordinates**

$$x = r \cos(\theta), \quad y = r \sin(\theta)$$

when we investigate limit problems. Notice that as $(x, y) \rightarrow (0, 0)$, we must have $r \rightarrow 0$.

Example 10. Find $\lim_{(x,y) \rightarrow (0,0)} \frac{e^{-x^2-y^2} - 1}{x^2 + y^2}$

Example II. Find $\lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2) \ln(x^2 + y^2)$

Remark. Consider

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + y^4} \quad (*)$$

By switching to the polar coordinates we obtain

$$\lim_{r \rightarrow 0} \frac{r \cos(\theta) \sin^2(\theta)}{\cos^2(\theta) + r^2 \sin^4(\theta)} \quad (**)$$

Clearly, the value of this limit is 0 for each **fixed** value of θ .

However, (*) dne as along the parabola $x = y^2$, (*) becomes

$$\lim_{y \rightarrow 0} \frac{y^2 y^2}{y^4 + y^4} = \frac{1}{2}$$

But along the x -axis the limit value is 0.

Additional Examples

Example 12. Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 \tan(y) - y^2 \tan(x)}{x^2 + y^2}$

Example 13. Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{x \sin(2y) - y \sin(2x)}{x^4 + y^4}$

Example 14. Find the value of α so that

$$f(x, y) = \begin{cases} \frac{(x^2 - y^2)^2}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0) \\ \alpha - 9, & \text{if } (x, y) = (0, 0) \end{cases}$$

is continuous at $(0, 0)$.

Example 15. Evaluate $\lim_{(x,y) \rightarrow (-3,2)} \frac{2x^2 + 5xy + 3y^2}{2x + 3y}$

Example 16. Evaluate $\lim_{(x,y) \rightarrow (1,0)} (1 + xy)^{1/xy}$

Example 17. Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{(x^2 + y^2)^2}{x^4 + y^4}$

Example 18. Find $a, b \in \mathbb{R}$, such that

$$f(x, y) = \begin{cases} \frac{x^2(y-1)}{3x^2 + 7(y-1)^2} + a, & \text{if } x > 0, \\ \frac{\tan^2(x+y-1)}{(x+y-1)^2} + b, & \text{if } x < 0, \\ 9, & \text{if } x = 0, \end{cases}$$

is continuous at the point $(0, 1)$.

Solution. For simplicity, let us make a change of a variable: $Y := y - 1$. Then we have

$$f(x, Y) = \begin{cases} \frac{x^2 Y}{3x^2 + 7Y^2} + a, & \text{if } x > 0, \\ \frac{\tan^2(x+Y)}{(x+Y)^2} + b, & \text{if } x < 0, \\ 9, & \text{if } x = 0, \end{cases}$$

Clearly, we have

$$\lim_{(x,y) \rightarrow (0,1)} f(x, y) = \lim_{(x,Y) \rightarrow (0,0)} f(x, Y).$$

For $x < 0$, we have

$$\begin{aligned} \lim_{(x,Y) \rightarrow (0,0)} f(x, Y) &= \lim_{(x,Y) \rightarrow (0,0)} \frac{\tan^2(x+Y)}{(x+Y)^2} + b \\ &= \lim_{(x,Y) \rightarrow (0,0)} \frac{\tan^2(\theta)}{\theta^2} + b \quad (\theta = x + Y) \\ &= 1 + b \end{aligned}$$

On the other hand,
For $x > 0$, we have

$$\lim_{(x,Y) \rightarrow (0,0)} f(x,Y) = \lim_{(x,Y) \rightarrow (0,0)} \frac{x^2 Y}{3x^2 + 7Y^2} + a$$

Since

$$\left| \frac{x^2 Y}{3x^2 + 7Y^2} \right| \leq |Y| \rightarrow 0 \quad (\text{as } Y \rightarrow 0)$$

we conclude that

$$\lim_{(x,Y) \rightarrow (0,0)} f(x,Y) = \lim_{(x,Y) \rightarrow (0,0)} \frac{x^2 Y}{3x^2 + 7Y^2} + a = 0 + a = a.$$

In order for f to be continuous at $(x,y) = (0,1)$, we must have

$$\lim_{(x,y) \rightarrow (0,1)} f(x,y) = f(0,1)$$

$$\implies$$

$$1 + b = a = 9 \implies$$

$$a = 9 \quad \text{and} \quad b = 8.$$